

Effects of Load and Focus of Attention on Mechanical Parameters During Bench-Press Throw in Resistance-Trained Men

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Purpose: Power output is dependent on the load used during exercise such as bench-press throw (BPT). Attentional focus (external [EXT] vs internal [INT]) during exercise significantly modulates power performance. The purpose of the present study was to examine the effects of load and attentional focus on mechanical parameters during BPT. **Methods:** In a crossover study, 31 resistance-trained men (mean age 23.5 [3.0] y) performed BPT at 30% (light), 50% (moderate), and 70% (heavy) of 1-repetition maximum (1-RM) using an INT or EXT focus of attention in randomized order. A linear encoder was used to identify barbell vertical displacement, throw time, peak/average velocity, force, and power during the concentric lifting phase. **Results:** Statistical analysis revealed significant load $\times$ focus interaction effects for velocity and vertical displacement ($P \leq .045$; $0.66 \leq d \leq 0.89$). Post hoc analyses indicated significantly larger velocities and displacements at 30% and 70% of 1-RM in favor of EXT ($P \leq .038$; $0.79 \leq d \leq 1.13$) but similar values at 50% of 1-RM ($P > .05$). Furthermore, significant main effects of load were found for throw time, force, and power ($P < .001$; $4.20 \leq d \leq 14.0$). While time and force gradually increased with higher loads ($P < .001$; $1.45 \leq d \leq 14.0$), power output was larger at 50% compared with 30% and 70% 1-RM ($P < .001$; $3.09 \leq d \leq 7.07$), irrespective of attentional focus. **Conclusions:** The present findings indicated that practitioners may use EXT over INT attentional focus to enhance velocity and vertical displacement during BPT at light and heavy loads (ie, 30% and 70% 1-RM). At moderate loads (ie, 50% 1-RM), mechanical bench-press parameters appear to be less affected by attentional focus.

Keywords: ballistic exercise, external focus, internal focus, load–velocity profile

The production of high mechanical power output in the upper limbs is an essential prerequisite for successful performance in sports such as boxing or handball involving rapid upper-limb actions (eg, punching and throwing).^{1,2} For the development of strength and mechanical power in the upper limbs, the (free-weight barbell) bench-press exercise has become one of the most frequently used exercises during resistance-training programs.^{3,4} Traditionally, the bench-press technique is characterized by large accelerations particularly at the beginning of the concentric movement.^{5,6} For maximal power production, ballistic bench-press exercises (eg, bench-press throw [BPT]) have been shown to provide superior training stimuli compared with nonballistic exercises.^{7,8} For instance, mechanical bench-press parameters (ie, average/peak velocity and power, and average force) were significantly larger during the ballistic throwing technique compared with the nonballistic movement at 45% of the 1-repetition maximum (1-RM) in resistance-trained males.⁷ Furthermore, 12 weeks of BPT training significantly improved 1-RM and shot-put performance compared with a nonballistic bench-press training at 30% to 50% of the 1-RM in male university athletes.⁸ The deceleration at the end of the nonballistic bench-press technique could have been less specific to the shot-put performance.⁸

One factor that is considered important for power output is the load used.⁹ In fact, larger loads will result in slower maximal

movement velocities.^{9,10} Mechanical power defined as the product of force and velocity is maximal at a compromised level of force and velocity.¹⁰ Based on the force–velocity profile for a specific exercise, maximal mechanical power and the intensity (ie, load and velocity) at which it is produced can be calculated and used for training on an individual basis.^{9–12} In this regard, Cronin and Sleivert⁹ reported that the majority of research indicated maximal power output during bench press/BPT at loads ranging between 30% and 70% of the 1-RM. Recently, a meta-analysis of the literature indicated that light loads ($\leq 30\%$ of 1-RM) produced significantly larger mechanical power during BPT when compared with moderate ($>30\%$ and $<70\%$ of 1-RM) and/or heavy loads ($\geq 70\%$ of 1-RM).¹²

Further, there is scientific evidence that verbal instructions significantly affect mechanical variables of upper-limb exercises.¹³ For instance, mean and peak power were significantly larger during BPT at optimal loads for maximal power output (ie, 50 kg) in resistance-trained males, when participants were asked to lower the bar fast (ie, eccentric action) prior to the push movement (ie, concentric action) compared with medium or slow lowering velocities.¹³ In another study, Makaruk et al¹⁴ examined the effect of focusing on a target ahead (ie, external focus of attention [EXT]) versus focusing on the arms (ie, internal focus of attention [INT]) on underhand and overhand shot-put performance using a 4-kg metal shot in national-level athletes. Findings indicated that shot-put distances were significantly larger when using an EXT compared with an INT or a control condition.¹⁴ These results are in line with literature demonstrating that the concentration on the movement effect (ie, EXT) results in superior performance and motor learning (eg, ball throw and ball kick) compared with concentration on body movements (ie, INT).^{15,16} Calatayud et al¹⁷ reported that

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activation levels and contraction times of the pectoralis and triceps brachii muscles during rapid bench press at 50% of the 1-RM were not affected by focus of attention. Thus, the superiority of EXT over INT for mechanical power output in upper-limb exercises (eg, BPT) may depend on the loads used during exercise. However, and to the authors' best knowledge, there is no information available in the literature on whether the role of attentional focus on mechanical parameters during BPT is affected by the exercise load.

Therefore, the aim of this study was to examine the effects of load and attentional focus (EXT vs INT) on mechanical parameters (eg, power, force, and velocity) during BPT. With reference to the literature,^{10,12,14,17} lower movement velocities and higher force levels can be hypothesized during BPT when loads are increased during exercise (eg, 30%–70% of 1-RM). Further, mechanical performance measures during BPT will be enhanced when using EXT compared with INT instructions at light and/or heavy loads but not at moderate loads.

Methods

Participants

An a priori power analysis with an assumed type I error of 0.05, and a type II error rate of 0.20 (ie, 80% statistical power) was calculated for sample size estimation using G*Power (version 3.1).¹⁸ With reference to Soriano et al¹² and Chua et al,¹⁶ the analysis indicated that 30 subjects would be sufficient to observe large effects of load and small effects of focus of attention on BPT power output. For inclusion, participants had to be free of injury, pain-free during maximal lifting, and performing bench press exercises as part of their weekly training routines. A load corresponding to a ratio of bench-press 1-RM to body mass of 1.0 to 1.2 was suggested to indicate advanced strength levels in men.¹⁹ Accordingly, participants were included if they could lift at least their own body mass (ie, 1-RM: body mass ratio ≥ 1).²⁰ Finally, 31 resistance-trained men (age: 23.5 [3.0] y, body mass: 79.4 [7.4] kg, body height: 178.8 [5.5] cm, body mass index: 25.7 [2.8] kg/m²) with 5.0 (2.9) years of resistance training experience were recruited. Participants were informed verbally and in writing regarding the implications and potential side effects of participating in the experiment. Further, participants were asked to refrain from any strenuous activity 48 hours prior to testing. The study was conducted in accordance with the Declaration of Helsinki, confirmed by the Norwegian Center for Research Data (926377), and approved by the institutional ethical committee (22/06742-4).

Study Design and Experimental Procedures

In this study, a single-group, repeated measures, randomized cross-over study design was used to examine the effects of load (ie, 30%, 50%, and 70% of 1-RM) and attentional focus (ie, EXT and INT) on mechanical parameters during BPT in resistance-trained males. Participants were invited to attend the laboratory on 2 different occasions separated by 48 and 72 hours (ie, familiarization session and experimental session). At the beginning of each session, participants completed a 5-minute general warm-up using aerobic exercises (ie, jogging or cycling). The specific warm-up was conducted in a Smith machine (Pivot 680 L, Pivot Fitness) and comprised 10 bench press repetitions using the 6 kg barbell only, 10 repetitions at 26 kg, followed by 4 repetitions at 50%, and 2

repetitions at 75% of the self-reported bench-press 1-RM. Participants used their preferred grip- and feet-width, which were measured initially and then controlled before each attempt in all conditions and sessions.²¹ During the familiarization session, participants' nonballistic bench-press 1-RM was assessed, and subsequently, BPT was performed across a range of different loads (30%–70% of 1-RM) to get accustomed to proper lifting/throwing technique. Two to 5 attempts at each load were provided.¹³ During the experimental session, participants performed maximal effort (ballistic) concentric BPT in the Smith machine with different loads (30%, 50%, and 70% of the participants' 1-RM) using different foci of attention (ie, EXT vs INT) in randomized order. Previous literature indicated that this load range will most likely cover the optimal load for maximal power output during bench press/BPT.^{9,10} Approximately 10 seconds prior to the attempt, the participants received instructions to concentrate on the movement effect (ie, EXT) or on body movements (ie, INT) during exercise.^{15,16} More precisely, the EXT instruction was "During this trial, try to focus on accelerating the load plates as fast as possible," while INT was made out to "During this trial, try to focus on using your chest and triceps muscles as fast as possible."

Assessment of Bench-Press 1-RM

The bench-press 1-RM was determined on the Smith machine for each participant as a reference value to ensure individual load adjustments for BPT. High test–retest reliability was previously reported for bench-press 1-RM with an intraclass correlation coefficient of .997.²² Similar to the BPT technique, participants were horizontally lying in supine position on a bench grasping the bar of the Smith machine over the chest. Subsequently, the participants unlocked the barbell from the Smith machine, lowered the barbell in a controlled manner, lightly touching the chest, and pressed the load upward. The assessment followed the recommendations of the National Strength and Conditioning Association.²³ Verbal encouragement was provided throughout the testing protocol. The absolute and normalized (relative to body mass) maximal lifted load were used for further analysis.

Assessment of Mechanical Parameters During BPT

Mechanical parameters during BPT were assessed by means of a linear encoder system (Ergotest Innovation A/S). The linear encoder (resolution: 0.019 mm, sampling rate: 200 Hz) was attached to the barbell and used to identify the participants' force–velocity profile. Previous research indicated that the instrument is valid and reliable for the assessment of the mechanical parameters' average velocity ($r = .993$, coefficient of variation [CV] = 2.54%) and displacement ($r = .993$, CV = 1.92%).¹³ During testing, participants were horizontally lying in supine position on a bench holding the bar of the Smith machine in extended arm position over the chest. Subsequently, participants were instructed to lower the barbell, lightly touching the chest (sternum position) and immediately press upward exerting maximal voluntary velocity to the point of vertical projection (ie, barbell throw). A short rest (approximately 1 s) was performed after the eccentric and before the concentric muscle actions to prevent eccentric-concentric movement coupling. Participants had to "lower the barbell fast, but controlled, and lift the barbell as hard and as fast as possible." Two trials per condition (ie, combined load and attentional focus) were conducted with rests between trials of 1 minute

and rests between conditions of 3 minutes. Trials were rejected if the barbell bounced off the chest or if the barbell did not touch the chest during lowering (ie, visible distance of ≥ 2 cm above the chest). Further, trials were rejected if the hips lifted off the bench during exercise.¹³ Barbell peak velocity, average velocity, time to peak velocity, average force, average power, vertical displacement, and total time of the concentric throwing movement were analyzed using a commercial software (Musclelab, version 10.4.37.4073, Ergotest Innovation A/S). The mean of the 2 trials was used for further analysis. Further, mean power scores were independently plotted against the load for EXT and INT conditions, and the first derivative of the polynomial mean power–load relationship was used to estimate the individual load for maximal power output each.²⁴

Statistical Analyses

Data are presented as group mean values and standard deviations. Normal distribution of data was tested and confirmed using the Shapiro–Wilk test. A 3 (load: 30%, 50%, and 70% of 1-RM) $\times$ 2 (attentional focus: EXT and INT) analysis of variance with repeated measures on load and attentional focus was calculated to analyze the combined effects of load and focus of attention on mechanical parameters during BPT. Homogeneity of variance was examined using the Mauchly sphericity test for the factor load. In case of homogeneity's violation, the Greenhouse–Geisser correction was applied for further analyses. Post hoc tests with the Bonferroni-adjusted α were conducted to identify the comparisons that were statistically significant. Additionally, effect sizes were calculated by converting partial eta-squared to Cohen d .²⁵ Cohen d values were classified as small ($0.35 \leq d < 0.80$), medium ($0.8 \leq d < 1.5$), and large ($d \geq 1.5$).²⁶ The significance level was set at $\alpha = .05$. All analyses were performed using Statistical Package for Social Sciences (SPSS, version 25.0).

Results

Participants' bench-press 1-RM was 110.8 (15.5) kg corresponding to a 1-RM:body mass ratio of 1.3 (0.2). Estimated loads for maximal power output during BPT were 54.0% (2.9%) of 1-RM for EXT and 54.4% (3.7%) of 1-RM for INT ($P = .401$).

Descriptive and statistical results are summarized in Table 1. The statistical analysis revealed significant and small to medium interaction effects of load and attentional focus on peak velocity, mean velocity, and vertical displacement ($P \leq .045$; $0.66 \leq d \leq 0.89$). Post hoc analysis indicated significantly larger peak velocities for EXT compared with INT when using loads of 30% ($P = .038$; $d = 0.79$) and 70% ($P = .015$; $d = 0.94$), but not 50% of the 1-RM ($P = .814$; Figure 1). Further, mean velocity was larger for EXT compared with INT at 30% of 1-RM only ($P = .009$; $d = 1.01$, Figure 2). Vertical displacement of the barbell was significantly larger for EXT than INT at 30% ($P = .005$; $d = 1.12$) and 70% ($P = .004$; $d = 1.13$), but not 50% of 1-RM ($P = .081$; Figure 3). Additionally, large main effects of load were observed for time to peak velocity, average force, average power, and total time ($P < .001$; $4.20 \leq d \leq 14.00$; Table 1). Post hoc analysis showed that time to peak velocity, average force, and total time were significantly larger with increasing loads (ie, 70% > 50% > 30% of 1-RM; $P < .001$; $1.45 \leq d \leq 14.00$). Further, average power was significantly larger at 50% compared with 30% and 70% of 1-RM ($P < .001$; $3.09 \leq d \leq 7.07$).

Discussion

To the best of the authors' knowledge, this is the first study to examine the effects of load and attentional focus on mechanical parameters (eg, power, force, and velocity) during BPT in resistance-trained males. The main findings were that (1) time (ie, to peak velocity and total) and average force during BPT were gradually larger as loads increased from 30% to 70% of the 1-RM, (2) the largest average power output was achieved at 50% compared with 30% and 70% of 1-RM, and (3) EXT enhanced peak/average velocity and vertical displacement during BPT at 30% and/or 70% but not at 50% of 1-RM when compared with INT.

Bench press and BPT are frequently used in resistance training programs for improving mechanical power and sport-specific performance (eg, shot-put distance and handball throwing velocity) in the upper limbs.^{8,27} Accordingly, the external load used during training appears to be an important determinant as it directly affects the individual force–velocity profile of the exercise.^{9,10} In fact, the present study showed that the force output and the total time were larger as the load during BPT was increased from 30% to 70% of 1-RM. This finding is well in line with the literature demonstrating the relationship between force and time/velocity as a function of load.^{10,13,28} For instance, Newton et al²⁸ examined the effects of different loads (ie, 15%–100% of 1-RM) on mechanical parameters (eg, force and total time) during BPT with or without countermovement in resistance-trained males. In their study, force output (531 vs 752 N) and time (545 vs 1333 ms) were gradually larger as load increased from 30% to 70% of 1-RM for BPT without countermovement.²⁸ Given that mechanical power is calculated as the product of force and velocity, it is maximized at a load corresponding to a compromised level of maximal force (ie, high loads) and maximal velocity (ie, light loads).¹⁰ Accordingly, we found the largest BPT power output when using moderate loads of 50% of 1-RM compared with light (ie, 30% of 1-RM) or heavy loads (ie, 70% of 1-RM). This result was substantiated by estimating the individual load for maximal power output. In the present study, the estimated load for maximal power was 54.0% of 1-RM (≥ 59.8 kg) for EXT and 54.4% of 1-RM (≥ 60.3 kg) for INT. Similarly, evidence indicated that maximal power during BPT mostly occurs at loads ranging between 30% and 70% of the 1-RM.⁹ In contrast to previous review articles^{9,10} and to the findings of our study, a systematic review with meta-analysis revealed that light loads ($\leq 30\%$ of 1-RM) produced significantly larger effects on BPT power output (ie, peak and average power) compared with moderate ($>30\%$ and $< 70\%$ of 1-RM) and/or heavy loads ($\geq 70\%$ of 1-RM).¹² However, it has been noted that power is dependent on factors such as the individuals' physical characteristics (eg, strength level) or the exercise/training mode (eg, barbell bounce technique).^{10,12,13,29} For instance, professional rugby league players were stronger in bench-press exercise (1-RM: 134.8 kg vs 111.0 kg) and reached maximal power at significantly lighter loads (51.1% vs 54.9% of 1-RM) when compared with college-aged rugby players.²⁹ Further, average power at loads ranging from 30 to 60 kg increased by 8% to 14% if the participants were instructed to bounce the barbell off the sternum compared with lower the barbell lightly touching the chest.¹³ Therefore, these factors could partly explain the discrepancies between our study and the meta-analytical findings of Soriano et al.¹² Of note, the authors specifically highlighted that the heterogeneity and the small number of the included studies should be considered as a limitation of their meta-analysis.¹²

Table 1 Mechanical Parameters of Bench-Press-Throw Exercise by Load and Attentional Focus in Resistance-Trained Males

	30%			50%			70%			Load		Attentional focus		Load × attentional focus	
	INT	EXT	Δ , %	INT	EXT	Δ , %	INT	EXT	Δ , %	P	d	P	d	P	d
Peak velocity, m/s	1.80 (0.19)	1.85 (0.15)	2.5	1.31 (0.14)	1.30 (0.11)	−0.4	0.82 (0.10)	0.84 (0.10)	2.4	<.001	14.77	.150	0.54	.045	0.66
Average velocity, m/s	0.98 (0.11)	1.01 (0.08)	3.7	0.75 (0.07)	0.76 (0.06)	1.1	0.52 (0.05)	0.52 (0.06)	1.3	<.001	13.33	.032	0.82	.010	0.89
Time to peak velocity, milliseconds	356.5 (47.3)	347.1 (39.1)	−2.6	462.7 (54.9)	465.3 (51.7)	0.6	680.6 (105.1)	679.2 (89.6)	−0.2	<.001	9.11	.598	0.19	.270	0.42
Average force, N	375.5 (54.9)	377.3 (54.4)	0.5	604.9 (85.5)	609.0 (91.0)	0.7	809.4 (116.1)	809.6 (115.9)	0	<.001	14.00	.194	0.49	.418	0.31
Average power, W	367.8 (71.6)	381.6 (62.4)	3.7	456.0 (84.1)	460.1 (76.1)	0.9	418.2 (77.5)	423.3 (77.2)	1.2	<.001	4.20	.099	0.62	.091	0.58
Vertical displacement, cm	66.7 (9.9)	69.1 (9.7)	3.6	52.6 (6.6)	53.2 (6.5)	1.3	44.6 (5.2)	45.4 (5.5)	1.7	<.001	7.71	.001	1.29	.028	0.79
Time, milliseconds	682.6 (54.9)	680.8 (58.5)	−0.3	698.7 (59.1)	699.8 (58.8)	0.2	863.4 (111)	870.6 (88.2)	0.8	<.001	5.47	.628	0.18	.526	0.26

Abbreviations: *d*, Cohen *d*; EXT, external focus of attention; INT, internal focus of attention; *P*, alpha error. Note: Values are mean (SD) and percentage difference between INT and EXT (Δ).

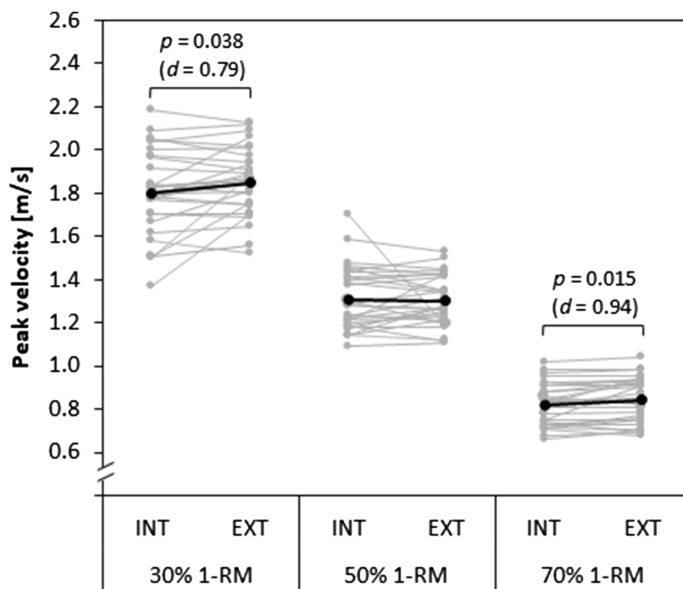


Figure 1 — Peak velocity during bench-press throw at different loads using an EXT versus INT in resistance-trained males. Gray dots/lines indicate individual scores, and black dots/lines represent mean scores. EXT indicates internal focus of attention; INT, internal focus of attention; 1-RM, 1-repetition maximum.

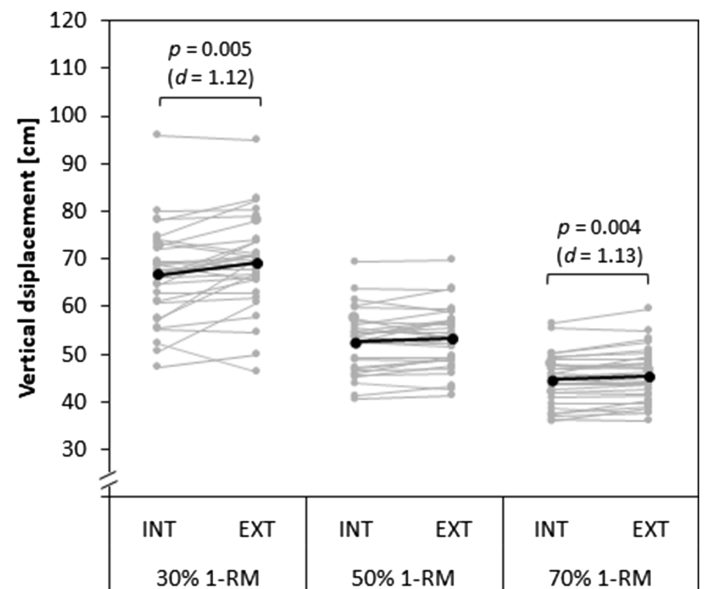


Figure 3 — Vertical displacement during bench-press throw at different loads using an EXT versus INT in resistance-trained males. Gray dots/lines indicate individual scores, and black dots/lines represent mean scores. EXT indicates internal focus of attention; INT, internal focus of attention; 1-RM, 1-repetition maximum.

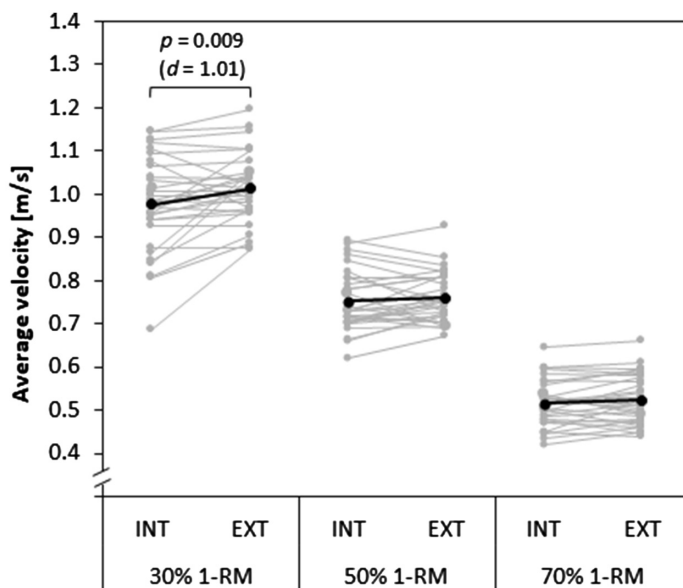


Figure 2 — Average velocity during bench-press throw at different loads using an EXT versus INT in resistance-trained males. Gray dots/lines indicate individual scores, and black dots/lines represent mean scores. EXT indicates internal focus of attention; INT, internal focus of attention; 1-RM, 1-repetition maximum.

More interestingly, the present study also demonstrated that the load used significantly modulated the effects of attentional focus on mechanical variables during BPT. Specifically, EXT enhanced peak and average velocity as well as vertical displacement at light (30% of 1-RM) and/or heavy loads (70% of 1-RM) only (2%–4%) when compared with INT. These small to medium effects (ie, effect sizes ranging from 0.79 to 1.13) appear to be even

greater than the trivial- to small-sized enhancements in BPT performance (effect sizes ranging from 0.11 to 0.43) induced by preceding conditioning activities (eg, bench press at 60%–84% of 1-RM, ballistic plyometric exercises) as reported in a recent systematic review with meta-analysis.³⁰ When using moderate loads corresponding to 50% of the 1-RM, focus of attention did not affect BPT mechanical parameters. There is abundance of evidence indicating that adopting an EXT strategy focusing on the movement effect rather than an INT approach focusing on the movement pattern resulted in enhanced performance and learning effects, irrespective of age, health status, or expertise level.^{15,16} As a potential explanation, it was suggested that INT causes individuals to constrain their motor system by interfering with automatic control processes, while EXT promotes a more automatic motor control strategy by utilizing unconscious, fast, and reflexive control processes (ie, *constrained action hypothesis*).¹⁵ In terms of upper-limb strength/power exercises, shot-put performance using a relatively light load (ie, 4-kg shot) was significantly larger in national-level athletes (1%–2%) when focusing on an external target (EXT) instead of internally focusing on the arm movement (INT) or an unspecific control condition.¹⁴ Further, EXT enhanced punching velocities by 4% and 2% compared with INT and control conditions, respectively, in trained combat athletes.³¹ In Olympic weightlifters, EXT instructions enhanced barbell velocities compared with INT instructions (15%) during heavy-load snatch lifts (ie, 80% of 1-RM).³² Therefore, our significant findings for beneficial effects of EXT on BPT velocity and vertical barbell displacement particularly at light (ie, 30% of 1-RM) and heavy loads (70% of 1-RM) appear to be well in line with the literature above and the constrained action hypothesis.

However, of special interest may be the question as to why mechanical parameters of BPT were not affected by attentional focus at 50% of the 1-RM. In this regard, activation levels and muscle contraction times of the pectoralis and triceps brachii muscles were previously investigated during rapid bench presses at moderate loads of 50% of the 1-RM in resistance-trained males

focusing on either the load (ie, EXT) or on the acting muscles (ie, INT).¹⁷ Interestingly, EXT and INT produced similar muscle activity and contraction times during exercise. The authors concluded that interference exists between rapid movement velocities and conscious motor control ensuring automatic control processes.¹⁷ In this regard, our study may indicate that automatic control processes during BPT were not affected by attentional focus specifically when using moderate loads close to the optimal load for maximal power output. However, when using light and heavy loads during upper-limb power exercises, motor control strategy and the corresponding mechanical characteristics appear to be compromised by INT compared with EXT instructions. Future studies may need to examine muscle and/or cortical activities to specifically elucidate the mechanisms behind the observed interaction of load–velocity profile and attentional focus during upper-limb power exercises (ie, BPT).

Practical Applications

The load used during power training is a practically relevant determinant for power output. Individualized training programs with loads for maximal power production have been recommended to maximize performance gains.^{9,11,24} Findings from the present study revealed that moderate loads should be used to induce the largest power output during BPT in resistance-trained males. Interestingly, the effects of load during BPT on performance interacted with instructions on attentional focus. More specifically, beneficial effects of EXT instructions compared with INT instructions for BPT kinematics at 30% and 70% of the 1-RM only indicated that strength and conditioning specialists should provide instructions on the movement effect during upper-body power exercises at light and heavy loads. In contrast, the focus of attention played only a minor role in rapid upper-limb muscle contractions if moderate loads close to the optimal load for maximal power output were used during power training. However, it has to be acknowledged that cross-sectional studies do not allow cause-and-effect relations. Longitudinal studies are needed to verify whether EXT and INT are similarly effective to improve performance following moderate-load, upper-limb power training in resistance-trained individuals. Further, in contrast to BPT using the Smith machine, free-weight BPT could have provided a more “natural” throwing technique.³³ However, it can be assumed that safety in the present study was enhanced during exercise due to the restricted movement (ie, using the Smith machine).³³ Additionally, a recent systematic review indicated that variation in measures such as movement velocity was less when using a Smith machine.³⁴ In support of this notion and with respect to the excellent reliability of the linear encoder used in the present study for assessing average velocity and vertical displacement (ie, $r = .993$), the observed changes were larger compared with the minimal detectable changes for average velocity (0.026 m/s) and vertical displacement (1.2–2.3 cm).

Conclusions

The present study demonstrated that moderate loads induced the largest power output during bench-press throw in resistance-trained males. Furthermore, EXT instructions enhanced peak and average velocity and vertical displacement during bench-press throw at light and/or heavy loads but not at moderate loads when compared with INT. When using light and heavy loads during upper-limb power exercises, motor control strategy and the corresponding

mechanical characteristics may be compromised by INT compared with EXT instructions.

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